Dr. D. Y. Patil Arts, Commerce & Science College, Pune

Department of Computer Science

Academic Year: 2025-26

S.Y.B.Sc.(Data Science) Sem-III

Subject: DS-203-MJ-P Lab Course on DS-201-MJ-T and DS-202-MJ-T

Data Structure-I Practical Assignment2 Solution

Date: 27/06/2025

Topic: Searching Techniques

1. Write a Python program to accept a value x from user and use Linear search algorithm to check whether the number is present in the array or not.

```
a=[1,2,3,4,5]
x=int(input("enter the number to search:"))
size=len(a)
flag=0
for i in range(0,size):
   if x==a[i]:
      flag=1
      break
if flag==1:
   print("Found")
else:
   print("Not found")
```

2. Accept a value x from user and use Linear search algorithm to check whether the number is present in the array or not and output the position if the number is present using python. (Using Function)

```
def Lsearch(a,x):
   flag=0
   for i in range(0,size):
```

```
        if x==a[i]:
            return i

a=[1,2,3,4,5]
x=int(input("enter the number to search:"))
size=len(a)

index=Lsearch(a,x)
if index or index==0:
    print("Found at",index)
else:
    print("Not found")
```

3. Write a Python program to accept a value x from user and use Binary search algorithm to check whether the number is present in the array or not.

```
a = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
x = int(input("Enter the value to search : "))

low = 0
high = len(a) - 1
flag=0
while low <= high:
    mid = (low + high) // 2

    if a[mid] == x:
        flag=1
        break
    if a[mid] < x:
        low = mid + 1
    else:
        high = mid - 1
if flag==1:
    print("Found")
else:
    print("Not Found")
```

4. Accept a value x from user and use Binary search algorithm to check whether the number is present in the array or not and output the position if the number is present using python. (Using Function)

```
def binarySearch(arr, x):
  low = 0
  high = len(a) - 1
  while low <= high:
    mid = (low + high) // 2

    if a[mid] == x:
      return mid

    if a[mid] < x:
      low = mid + 1
    else:
      high = mid - 1

  return -1

a = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
x = int(input("Enter the value to search : "))

result = binarySearch(a, x)

if result != -1:
 print("Found at index", result)
else:
 print("Not found")
```